MSR Data Challenge: GitSkills Checkpoint 2

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1 Initial Research and Research Questions

This checkpoint reports our progress on the MSR 2027 GitSkills mining challenge, which uses GitSkills, a July 2026 collection of public GitHub skill files in Anthropic's agent-skill format.[1] A skill file tells an agent how to carry out a task, and its instructions can name outside products, which is what this study counts. We work from release version 3 of 11 September 2026, and our code and notebooks are kept in the study repository.[2]

1.1 Research questions

Checkpoint 1 asked three questions, but two of them measured the same step, since the study cannot assign a license to a tool name until it knows the project that publishes that tool, so we merged those two and now ask the questions below.

- Which tools appear most often in skill files?
- How are the projects that publish those tools licensed, once each license is grouped by the freedom it gives?

1.2 Study population

To keep the population small and well defined, a skill stays in the study only when every rule below holds, whatever its body length, although the checks in this report used the full tables.

- The text is counted once, so copies of the same skill are left out.
- The header metadata at the top of the file is valid.
- The file is named `SKILL.md` or `skill.md`, and its body is plain text rather than JSON or CSS.
- The file sits in the usual skill folder under `.claude/skills` or in another skills directory.

1.3 Data and licenses

The collection has four tables, covering the skill files, the repositories that contain them, the files bundled with each representative skill, and notes on how the collection was gathered, while commit history is stored on a fetched skill row rather than in a table of its own. Extraction reads only the skill body, and because the license string on a skill's repository belongs to that hosting project, the second question needs the project that publishes each taught tool instead, while a named service with no such project stays unresolved.

2 Work Completed and Technical Details

2.1 Done so far

We downloaded the published release once and kept it as a local copy, so later updates to the public collection stay out of the analysis, and we analyze its tables with Python and Polars. Names are extracted with GLiNER,[3] and the finished work covers the items below.

[1] `https://arxiv.org/abs/2608.10906`
[2] `https://github.com/nichmorgan/msr-data-challenge-2027`
[3] `https://arxiv.org/abs/2311.08526`

- A frozen local copy of the collection.
- Checks of star counts, host-repository license labels, and the text stored beside a skill.
- A label-only model run on 500 companion texts.

Every notebook sets one shared random seed when it loads the tables, and the 500-text draw uses that seed, so a rerun returns the same texts, each still tied to the file it came from. The twelve-text trial in the last section drew its sample with its own fixed seed. Dated PDF exports of the explore notebooks are kept in `scripts/exports`, so their outputs can be read without running the notebooks, which may still change.

2.2 Score cutoff and accuracy

The model gives each hit a score, and a cutoff on that score decides which hits we keep. We are still studying how that cutoff relates to accuracy, and for now we test the simplest pairing, so a cutoff of 0.8 is checked against an accuracy near 80%, although that pairing is only an arbitrary starting point. To measure accuracy, the authors will label the taught tools in a random sample of the population, drawn with the shared seed and sized before anyone sees the model output. We will report precision, the share of extracted tools that are truly taught tools, and recall, the share of truly taught tools the model finds, along with inter-rater agreement on a subset that two authors label independently, but no such number appears here because that sample has not been labeled.

3 Pilot Findings

3.1 License labels and companion text

After trimming, lowercasing, and filling blank labels, the repository table held 42 distinct license strings, among them MIT, Apache-2.0, GPL-3.0, and AGPL-3.0. These strings describe the repository that hosts a skill, so they are a data check rather than an answer about the projects that publish taught tools. We also classified the 3,486,179 non-blank texts stored beside a skill, and Table 1 shows that almost all of them are free text or JSON objects. Both kinds include scripts and configuration, so companion text is not skill prose, and extraction stays on the skill body.

Table 1: Kinds of non-blank text stored beside a skill.

Kind	Rows	Share (%)
Free text	3,236,628	92.8
JSON object	238,834	6.9
JSON array	10,523	0.3
Other JSON[4]	194	0.0

3.2 Label-only model run

Before we settled on the skill body, a seeded draw of 500 companion texts, each clipped to its first 384 characters, was passed through the label-only model at a score cutoff of 0.8. Figure 1 shows every span found in at least two texts, and no bar comes near 500 because the cutoff drops every lower-scoring span, so many texts return no span at all. The chart also shows why a human check is needed, since the skill filename is marked as a tool even though it names the document itself, and GitHub, which leads with nine files, can be the site inside an address, the place the skill is published, or an attribution link, while none of these labels has a measured error rate.

[4] A JSON value that is a string, a number, a boolean, or null.

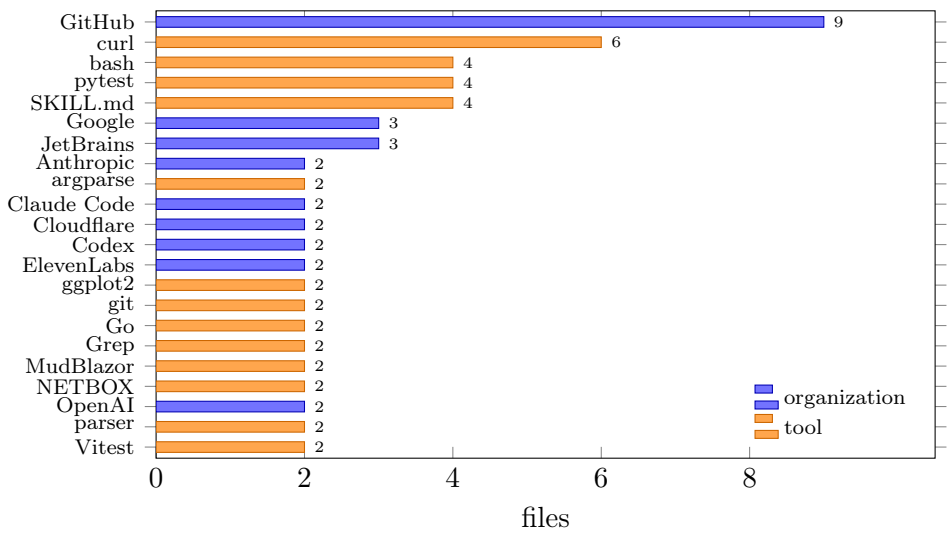


Figure 1: Spans that the label-only model found in at least two of 500 companion texts at a score cutoff of 0.8, counted as distinct files after folding case. Bars stay small because many texts return no span above the cutoff.

4 Remaining Work and Responsibilities

4.1 Relation model

Relation extraction is still in progress, because we think a link between a tool and the organization that publishes it can filter false positives such as the skill filename or GitHub inside an address, so we are looking for entity and relation labels that make that filter possible. Our first try ran GLiNER Relex large, version 0.5,[5] on the same 500 companion texts at a cutoff of 0.6, and while it linked a tunnel setting to Cloudflare as we would want, it also tied `python3` to a nearby word and a copied repository address to GitHub. A second trial on twelve short skill texts, whose length limit applies only to that trial, still links the wrong pairs, and since we read those rows manually, they show where extraction fails but not how often.

4.2 Next steps

Because an extraction rule settled too early would count the wrong tools for the first question and send the wrong set into the second, the remaining steps run in the order below, which the December 3 presentation and the December 9 paper also follow.

- Choose entity and relation labels that filter false positives.
- Label the validation sample, and record precision, recall, agreement, and the errors.
- Extract tools from the full population.
- Find the repository for each confirmed tool, read its license, and group licenses by freedom.
- Answer the two questions.

Each step in this list appears in the future work section of the December 9 paper, and the limitations section of that paper names what these steps have not settled yet, such as the unlabeled validation sample and the labels still to be chosen. When a step is finished, we move it into the method and results of the paper and remove the limitations it resolves, so both sections shrink as the study advances. One limitation is meant to stay, because we validate a random sample rather

[5] Model card at `https://huggingface.co/knowledgator/gliner-relex-large-v0.5`.

than every row, and errors in the rows nobody reviews remain in the counts.

4.3 Responsibilities

Aidan wrote the questions, the related work, an early look at the data, and the first load of the tables, while Samuel set up the shared Python environment and wrote the project plan. Morgan wrote the method, the check on model scores, and the local copy of the collection, and all three of us share the remaining analysis, the validation sample, and both December deliverables.